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Section A: Objective Questions (15 Marks)

MCQ: 1 mark each | MSQ: 2 marks — 0 marks if not all correct options are selected

Q1. A Capgras patient can correctly identify a familiar person but reports that the person "feels unfamiliar" and is therefore believed to be an imposter. This pattern most strongly suggests:
(Select ALL correct options — 2 marks, 0 if incomplete)

(Select ALL that apply — 2 marks, 0 if incomplete)

- ☐ A) Familiarity judgments depend solely on perceptual identification
- ☒ B) The affective component of recognition contributes to beliefs about identity
- ☒ C) There is a disruption in the autonomic/emotional response to familiar stimuli
- ☒ D) A functional disconnection exists between face recognition areas and limbic structures
- ☐ E) The deficit reflects a generalized memory impairment
- ☒ F) The ventral visual pathway supporting face recognition is intact

Q2. Which anatomical term refers to the "deep grooves" or "valleys" on the surface of the cerebral cortex?

- ☒ A) Fissures
- ☐ B) Gyri
- ☐ C) Axons
- ☐ D) Nodes

Q3. According to the principle of localisation of function, the occipital lobe is primarily associated with:

- ☐ A) Motor planning
- ☐ B) Auditory processing
- ☒ C) Visual processing
- ☐ D) Touch processing

Q4. What is a likely cognitive disorder where individuals struggle to recognize faces or interpret facial expressions despite having normal vision? (Determine by elimination)

- ☒ A) Prosopagnosia
- ☐ B) Parkinson's Disease
- ☐ C) Wernicke's Aphasia
- ☐ D) Capgras Syndrome

Q5. In the hierarchical organization of the brain, which structure is part of the "Reptilian complex" (Triune brain theory) responsible for basic physiological functions?

- ☐ A) Amygdala
- ☐ B) Limbic System
- ☐ C) Neocortex
- ☒ D) Cerebellum (Core structures)

Q6. Which Gestalt principle describes the tendency to perceive objects that are close to each other as forming a group?

- ☐ A) Closure
- ☒ B) Proximity
- ☐ C) Continuity
- ☐ D) Similarity

Q7. Which of the following most likely involves central control from the motor cortex?

- ☐ A) Knee jerk response when hit with a rubber hammer
- ☒ B) Walking in the garden
- ☐ C) Eye blink response when air is blown into eyelids
- ☐ D) Balancing on an unstable platform

Q8. A researcher asks participants to describe their thoughts and feelings while solving a puzzle. Different participants give inconsistent descriptions, and some report they "just knew the answer" without being able to explain how. What is the main limitation highlighted?

- ☒ A) Subjectivity and unreliability of introspection
- B) Absence of reinforcement
- C) Lack of ecological validity
- D) Overemphasis on behaviour

Q9. A 55-year-old patient can initiate movements but shows difficulty in tasks like buttoning a shirt or writing. Movements lack smoothness and precision during complex motor activities. Which brain area is most likely affected?

- A) Primary motor cortex (M1)
- B) Cerebellum
- C) Prefrontal cortex
- ☒ D) Basal ganglia

Q10. Which of the following would NOT be compatible with introspection?

- A) Describing subjective feelings
- B) Analysing conscious experience
- C) Reporting one's own thoughts
- ☒ D) Measuring reaction times

Q11. In the Motor System, which brain area is specifically associated with the coordination of complex motor patterns?

- ☒ A) Cerebellum
- B) Primary Motor Cortex (M1)
- C) Basal Ganglia
- D) Prefrontal Cortex

Q12. Which finding best supports Tolman's theory over strict behaviourism?

- A) Behaviour is entirely stimulus-driven
- B) Responses are fixed patterns
- C) Learning occurs only with reinforcement
- ☒ D) Animals can learn without immediate reward

Q13. Which psychological movement rejected introspection and insisted that psychology should focus only on mechanisms and processes that are objective and observable?

- ☒ A) Behaviourism
- B) Gestalt Psychology
- C) Cognitivism
- D) Structuralism

Q14. Which of the following best describes the core deficit in Capgras syndrome?

- ☒ A) Lack of emotional familiarity despite recognition
- B) Loss of long-term memory
- C) Inability to recognise faces
- D) Inability to see faces

Section B: Short Answer Questions (5 Marks)
(2.5 marks each)

Q1. Explain how bottom-up and top-down processes interact in perception. Illustrate your answer with an example.

Q2. Why is it that neural transmission is slow as compared to electrical transmission? *we get used over morning to recognize the words*